Electronics Research Institute

Crop Yield Prediction Using Decision Tree and Random Forest

Presented to Dr. Heba Draz, ECIP Internship Program Committee
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Course: Hardware/Software Co-Design
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Agenda

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Agenda

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Problem Statement& Objective

Why is it important to fix?

Problem Statement & Objective

Crop yield depends on many interacting factors

For example: soil moisture, temperature and sunlight hours.
Farmers usually predict crop yield using a mix of experience and observation.

What about using magic?

- Benefits:
  - Faster and more precise than traditional guesswork.
  - Helps in planning harvests, marketing, and resource allocation.
  - Can adapt to climate change scenarios.

- Machine learning is the modern “magic” that turns data on weather, soil, and crops into accurate yield predictions.

About the Authors

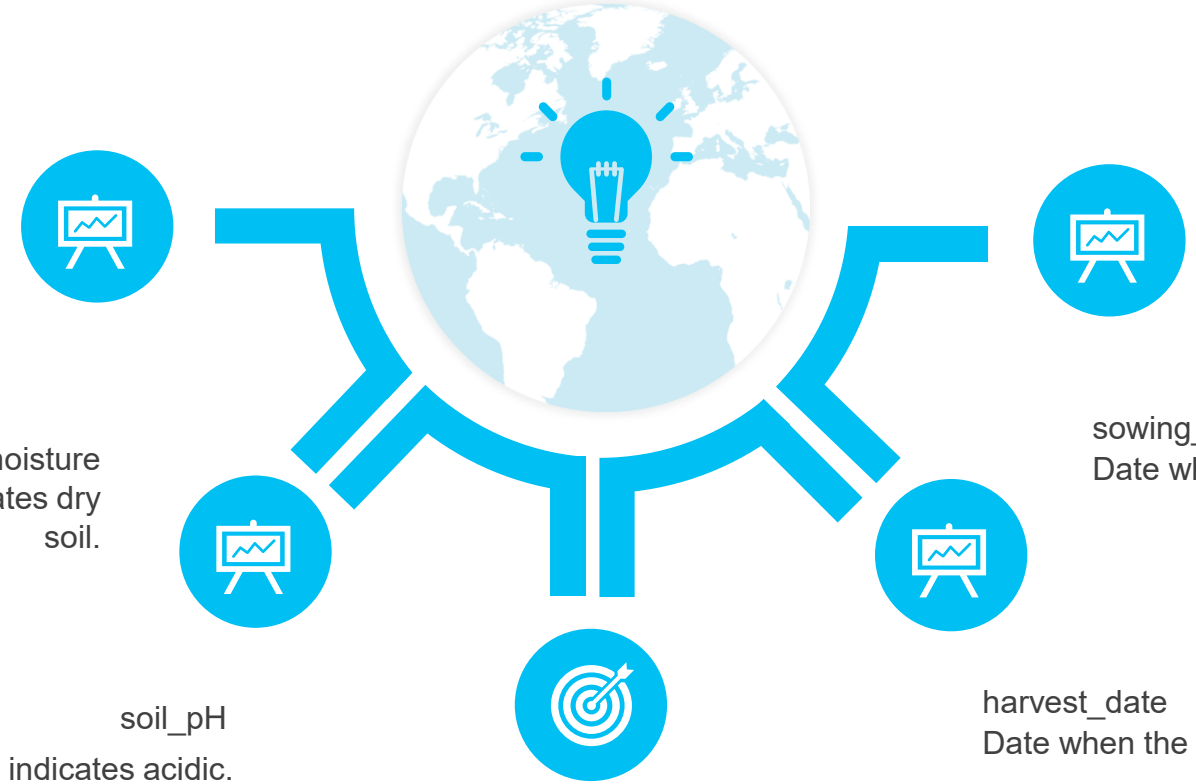
Dataset Title:

Smart Farming Sensor Data for Yield Prediction

Dataset: Author (Atharva Soundankar)
https://www.kaggle.com/datasets/atharvasoundankar/smart-farming-sensor-data-for-yield-prediction/data

Source Code: Author (ADITYA VERMA)
https://www.kaggle.com/code/adityaver/unit-5

Dataset Overview



Soil moisture
High indicates wetter soil, low indicates dry soil.

sowing_date
Date when the crop was sown.

soil_pH
High indicates alkaline, low indicates acidic.

harvest_date
Date when the crop was harvested.

yield_kg_per_hectare
Crop yield per hectare (target variable)

Set up ML environment

Model Evaluation Metrics

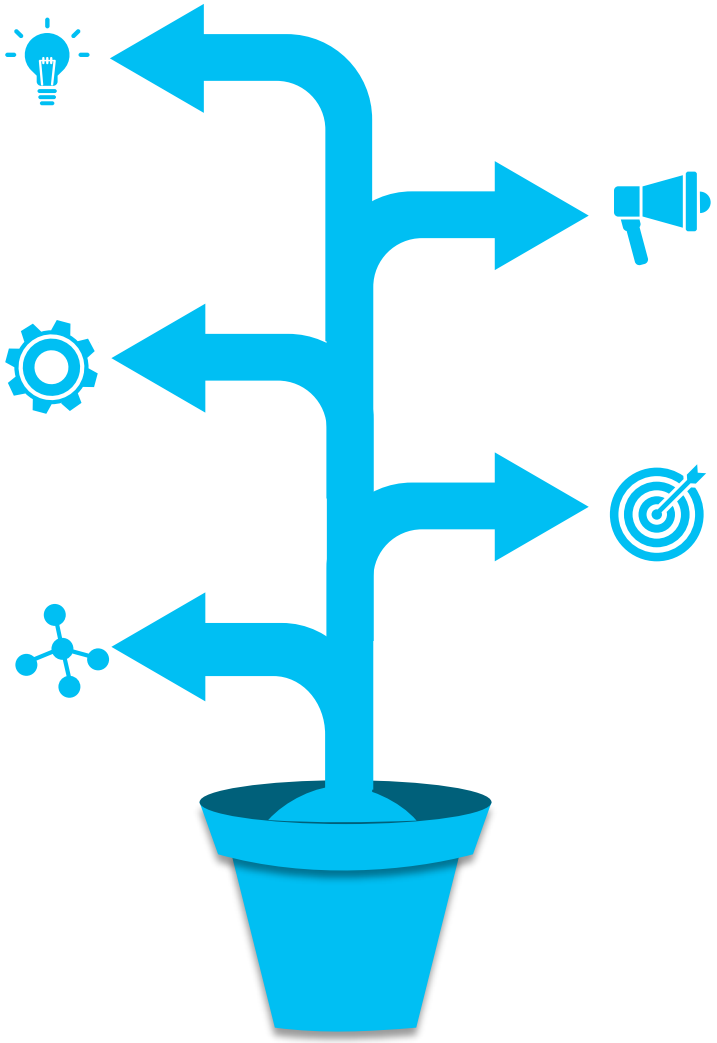
- Mean Squared Error (MSE)
- R^2 Score

Decision Tree Regressor

- Tree-based model using if–else rules
- Easy to interpret
- Can overfit the data

Data Preprocessing

- Handling missing values using median
- Date conversion to datetime format
- Train-test split (80% training, 20% testing)



Random Forest Regressor

- Ensemble of multiple decision trees
- Reduces overfitting
- More robust to noise
- Better generalization

Feature Engineering & Scaling

- Created new feature: Growing Days
- Calculated as harvest date minus sowing date.
- Feature scaling using StandardScaler.
- We need to scale features in ML because many algorithms are sensitive to the range and distribution of input values.
- Xscaled =X−mean(X) / std(X)

Machine Learning Algorithms

Decision Tree

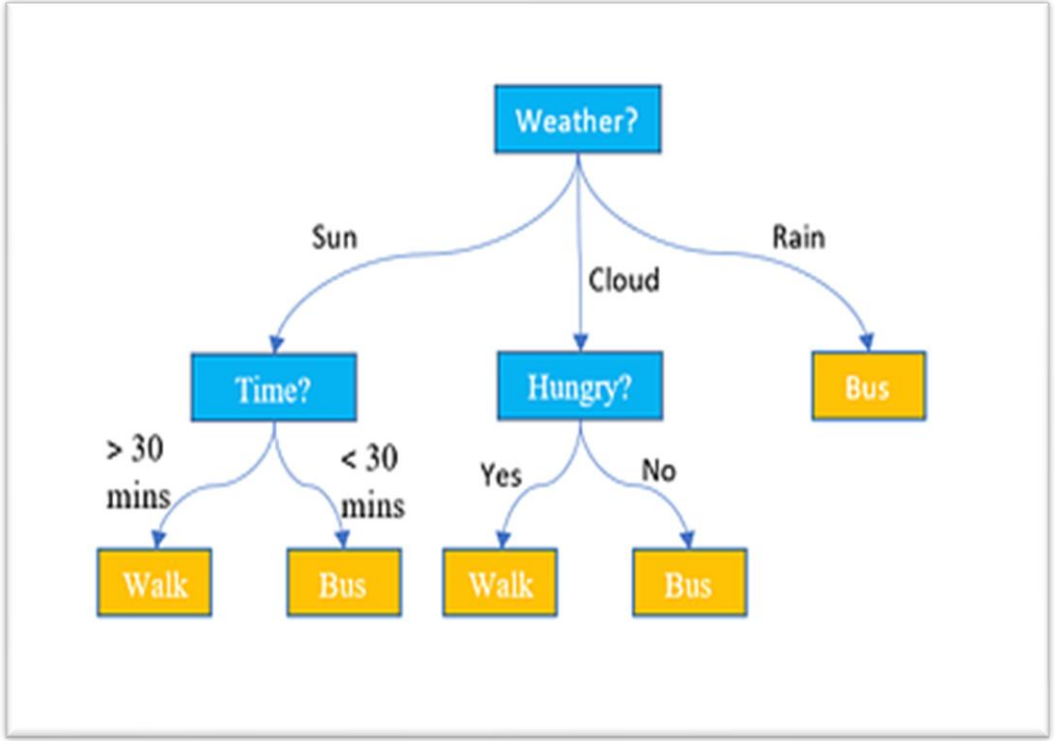
DT A supervised machine learning algorithm used for classification and regression.

How it works?

- Start at the root node with all data.
- Split data using the feature that best separates classes (or minimizes error).
- Repeat recursively to form branches until stopping criteria (max depth, min samples) are met.
- Leaf nodes give the final prediction.

Key Features:

- Simple to understand and interpret.
- Can handle numerical and categorical data.
- Prone to overfitting if the tree is too deep.



Random Forest

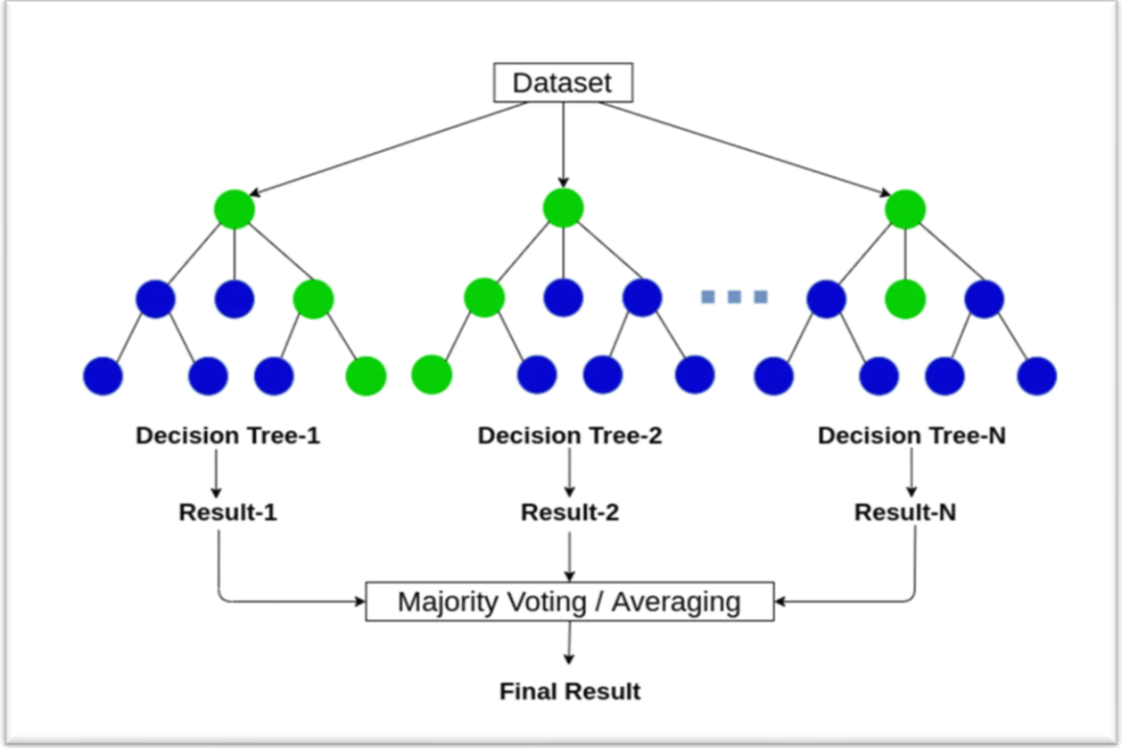
RF An ensemble machine learning method that combines many decision trees.

How it works?

- Randomly sample data
- Grow multiple decision trees using random feature subsets.
- Aggregate predictions:
  - Majority vote for classification
  - Average for regression

Key Features:

- Reduces overfitting compared to a single decision tree.
- Handles both classification and regression tasks.
- Robust to noise and missing values.



Model Evaluation Metrics

Model Evaluation Metrics

1 Mean Squared Error (MSE)

MSE is a regression metric that measures the average of the squared differences between the actual values and predicted values.

$$MSE = \frac{1}{n} \Sigma \underbrace{\left(y - \hat{y} \right)^2}_{\text{The square of the difference between actual and predicted}}$$

Key feature: ?

2 R^2

- Measures how well your model explains the variance in the target variable.
- Values range from 0 to 1 (sometimes negative if the model is worse than predicting the mean).

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}$$

Key feature: ?

Results & Comparison

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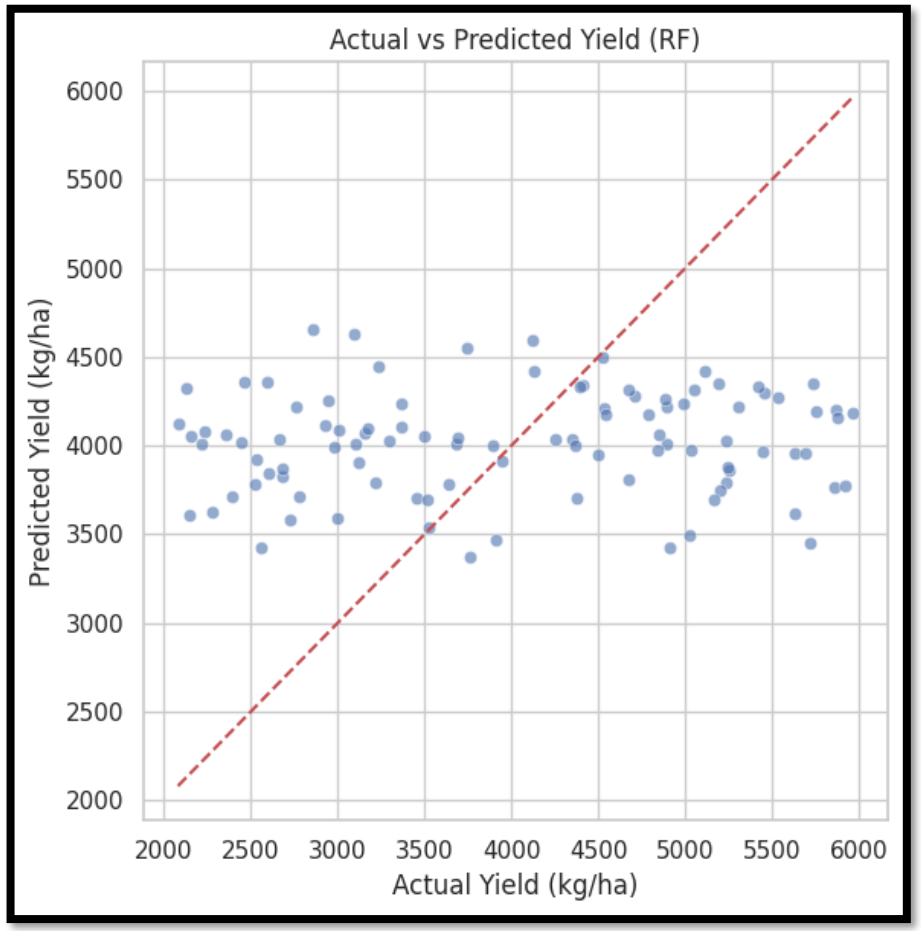
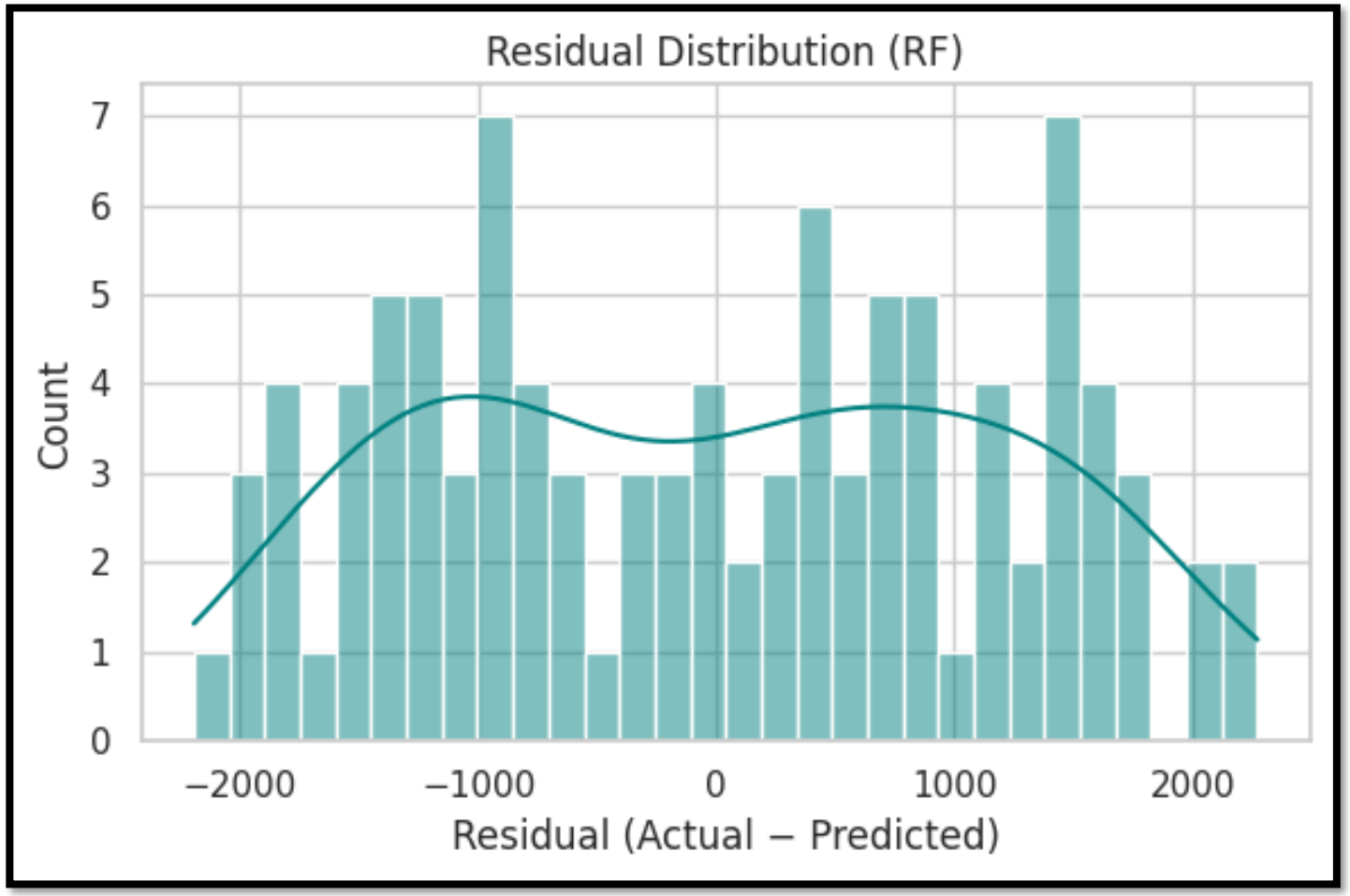


Results & Comparison



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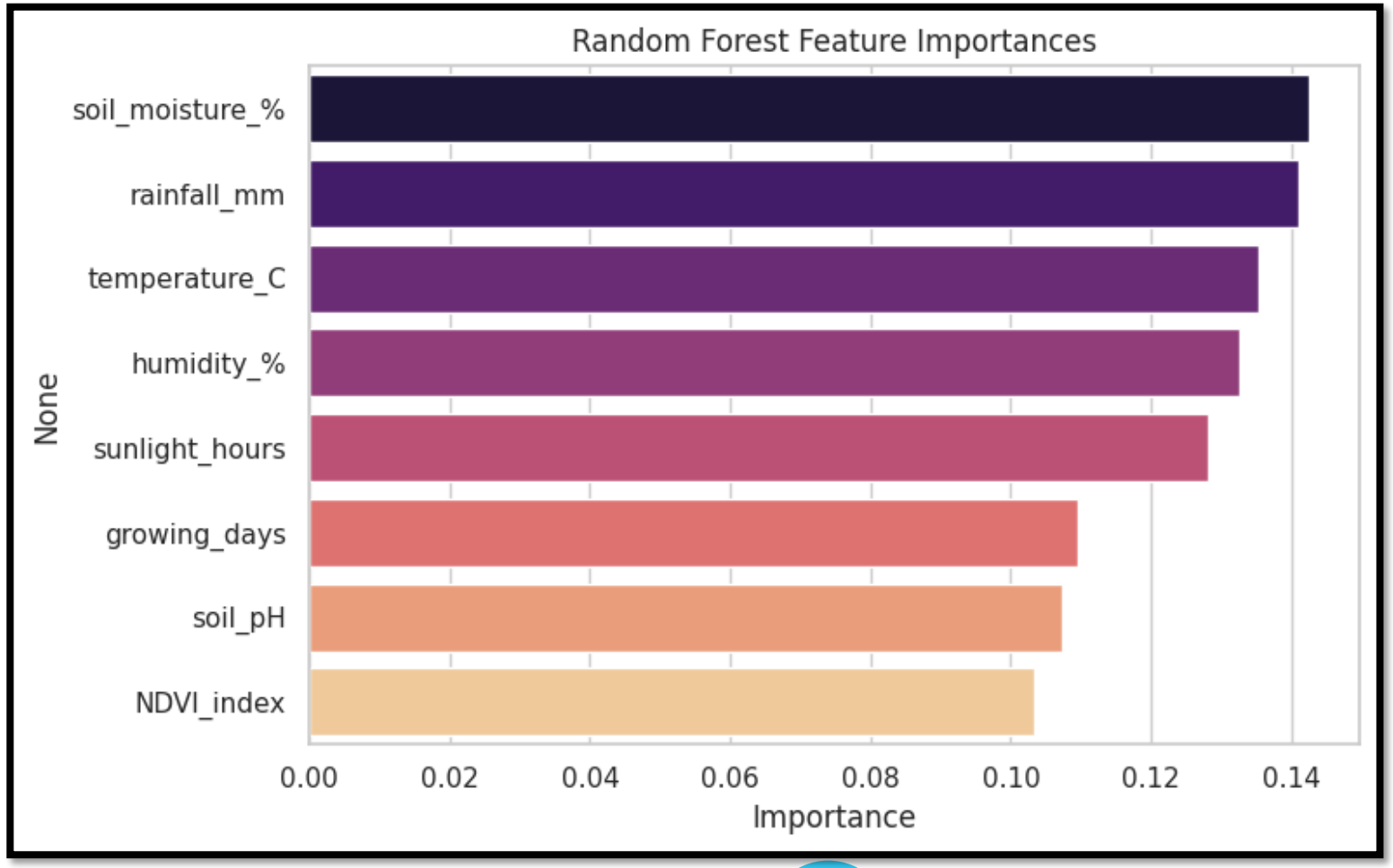
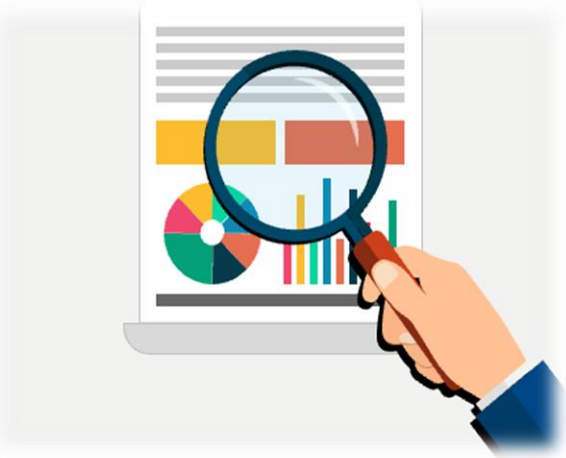


Results & Comparison

1 Random Forest achieved higher R^2 and Lower MSE

2 Random Forest generalized better



DT Cross-Validation (CV) R^2: -0.154 ± 0.084
RF Cross-Validation (CV) R^2: -0.065 ± 0.070
Decision Tree Test MSE: 1682896.19, R^2: -0.219
Random Forest Test MSE: 1414056.92, R^2: -0.024

Results & Comparison

```
Root Split:
• Soil moisture ≤ 0.49
  ├─ Low sunlight → Lower yield (~3200–4700)
  └─ High sunlight → Yield depends on pH, rainfall, NDVI

• Soil moisture > 0.49
  ├─ Low NDVI → Low yield (~2900)
  └─ High NDVI → Yield depends on rainfall & temperature
```

Conclusion

STRENGTHS

- Well-organized and structured dataset
- Suitable for FPGA-based implementation

WEAKNESS

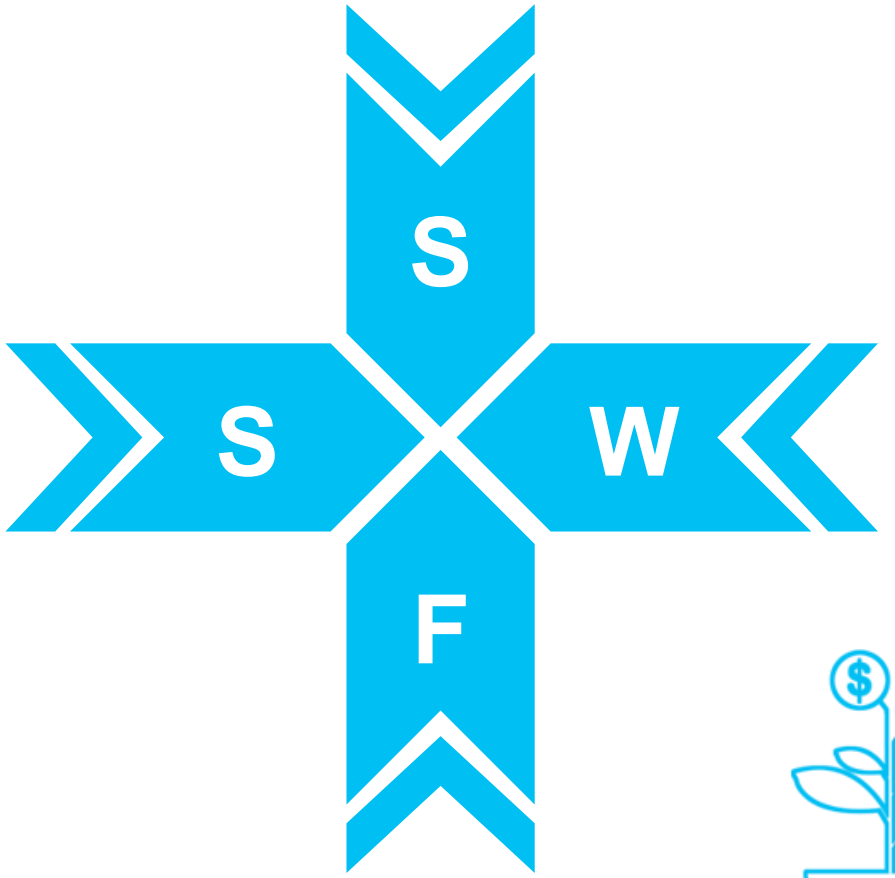
- Model performance depends strongly on data quality

Future Work

- Finalize ML environment
- Prepare and implement model on FPGA

Summary

- Built and evaluated Decision Tree and Random Forest models for crop yield prediction



References

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02 https://mlkeww00jjj8.i.optimole.com/cb:_P4T.27e/w:380/h:422/q:mauto/f:best/https://www.displayr.com/wp-content/uploads/2018/07/what-is-a-decision-tree.png

03 Dataset:https://www.kaggle.com/datasets/atharvasoundankar/smart-farming-sensor-data-for-yield-prediction/data

04 Code:https://www.kaggle.com/code/adityaver/unit-5

05 https://en.wikipedia.org/wiki/Random_forest

Thank you

Any Questions?